Nuclear Hazards

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Nuclear hazards are caused by radioactive substances that pose a risk to human health/environment. These radioactive substances can be either naturally occurring or man-made.

Natural Sources of Radiation

Cosmic radiation from outer space, quantity depends on altitude and latitude (higher at higher altitudes and latitudes). Terrestrial radiation: Emissions from radioactive elements from the Earth's crust, e.g., radon-222, soil rocks, air, water and food containing one or more radioactive materials. Internal radiation: All people contain radioactive potassium-40, carbon-14, lead-210 and other isotopes inside their bodies from birth.

Anthropogenic Sources of Radiation

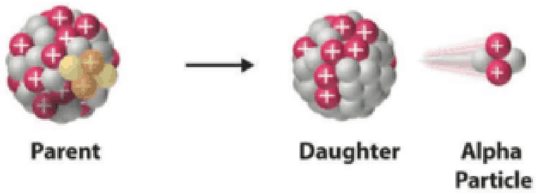
Some of the anthropogenic sources of radiation include:

1. Radiation emitted during the mining and processing of radioactive ores.
2. Radioactive materials in nuclear power plants and reactors, both raw materials and nuclear waste.
3. Radioactive fallouts during nuclear weapons testing, both on the surface, and in the oceans can also release a large amount of radiation. So too can leaks/accidents in nuclear power plants.
4. Radioactive isotopes in medical technology (x-ray machines, radioisotopes used in medicine) also release radiation.

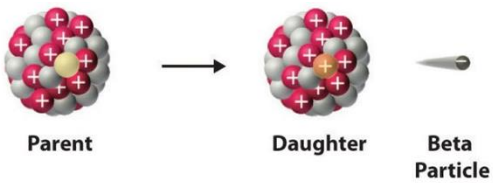
Types of Ionizing Radiations Emitted

When a radioactive nucleus undergoes decay, it breaks up into several smaller atoms, also called daughter isotopes, along with various types of ionizing radiations, which can be divided into three types.

Alpha particles: are released when a radioactive nucleus breaks up into daughter isotopes and positively charged helium atoms, also called alpha radiations. These are inherently positively charged. They have low penetrating power, and can be easily blocked by a sheet of paper or cardboard. They move relatively slowly, and are attracted by negative charges.



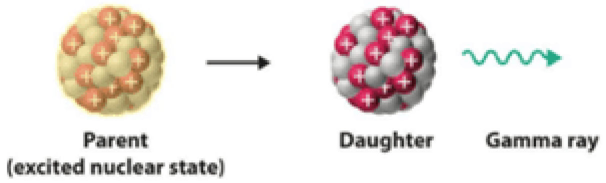
$$ {}^{A}_{Z}X \rightarrow {}^{A-4}_{Z-2}Y + {}^{4}_{2}He $$



$$ {}^{A}_{Z}X \longrightarrow {}_{Z+1}^{\ \ A}X' + {}^{\ 0}_{-1}\beta $$

Beta particles: on the other hand, are produced when a parent isotope undergoes decay into daughter isotopes, along with the release of electrons, also called beta radiations. These are usually high-speed electrons with high penetrating power, requiring a thin aluminium plate to stop. Being electrons, they are negatively charged, and are deflected by positive charges.

Gamma rays: are photons that move at the speed of light. They are electromagnetic waves and can be blocked only a thick lead or concrete block. They are not affected by electric charges, either positive or negative, and are neutral in nature.



$$^{Am}_{Z}X \rightarrow {}^{A}_{Z}X + {}^{0}_{0}\gamma$$

Effects of Radioactive Poisoning

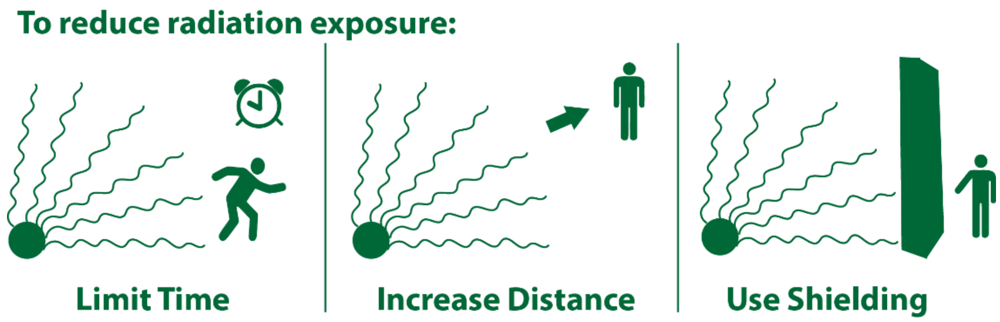
Radioactive emissions can penetrate biological tissues. For this reason, radiation is used to destroy cancerous tumours. But high levels of radiation > 100 rem can cause cell division blockage, prevents the normal replacement of blood, skin and other tissues, resulting in radiation sickness, and finally death.

- Very high doses of radiation may totally destroy cells, causing immediate death
- Lower doses may damage DNA, causing malignant tumours, cancers such as leukaemia. It also weakens the immune system, causing mental retardation and cataracts.
- DNA mutations affect genes and chromosomes, and are often carried over to offspring, up to several generations
- Acute exposure: burns and radiation sickness, burns, miscarriages, eye cataract, cancers of bone, thyroid, skin, lungs etc.

Protection from Radioactive Pollution

In case of a leak, there are only three ways in which we can protect ourselves from radioactive pollution.

- **Exposure time:** the lesser the amount of time that you are exposed for, the lower the dose of radiation that you will receive.
- **Distance:** the farther away that you are from the source of radiation, the less intense its effects will be.
- **Shielding:** shielding yourself behind a thick concrete or lead door can stop most of the harmful ionizing radiation, since they are very good at withstanding penetration.



Case Study: Chernobyl Disaster

The Chernobyl disaster is considered the worst nuclear disaster in history both in terms of cost and casualties. It occurred on 26 April 1986 at the Chernobyl Nuclear Power Plant, near the city of Pripyat in the north of the Ukrainian SSR in the Soviet Union. It is one of only two nuclear energy accidents rated at seven–the maximum severity–on the International Nuclear Event Scale, the other being the 2011 Fukushima nuclear disaster in Japan. The accident was caused by the malfunctioning of one of the steam turbines destabilizing the reactor. But this risk was not made evident immediately. So, operators continued the testing. Instead of shutting down, an uncontrolled chain reaction began, resulting in a core melt down. Explosions ruptured the core, destroyed the building, resulting in an open-air reactor core fire, which released considerable radioactive contamination that was airborne, and spread for over 9 days, precipitating into other parts of the USSR and Western Europe, before ending on 4th May 1986. The reactor explosion killed two engineers immediately. Overall, 237 suffered from acute radiation sickness, of whom 31 died within the first three months. The most lethal radionuclides that spread from Chernobyl were iodine-131, caesium-134, caesium-137 and strontium-90. The initial emergency response, together with later decontamination of the environment, involved more than 500,000 personnel and cost an estimated 18 billion Soviet rubbles—roughly US$68 billion in 2019, adjusted for inflation.

The Chernobyl nuclear power plant in May 1986, a few weeks after the disaster